

Aerial Video Cheat Sheet:

The setup for Cinematic Drone Shots you'll want to watch on repeat



I'm showing you only what you need to know to get insanely pretty shots every time.

Drones:

Mavic 2 Zoom, Pro, Mavic Pro 1, Mavic Air 2, Phantom 3 line, Phantom 4 line

(Mavic Mini just for shutter/framing)

Any settings I adjust will have a similar appearance in your drone's software.

Equipment:

STRONGLY recommend an ND Filter, they are literally just sunglasses for your camera. 1 extra battery at least, and car charger.



This allows you to shoot with a slower shutter, which is the last secret trick to getting cinematic looking footage. Ask any Film Camera Operator you know.

It's easy to find on Amazon.

For most situations, 16 ND will do the trick. For total broad daylight you might need a 32 ND. I have both, since they are only [\\$16 each for the Mavic Pro 1](#)

You can find it on amazon for \$40 for Mavic 2 Pro

[SkyReat is a good brand](#)

*NOTE: These are ND FILTERS (Neutral Density) *not* POLARIZER FILTERS. Those do a different thing and are the go-to suggestion from every novice with a drone, but are not worth the boat-load of money.

Beyond a 2nd battery just incase, you don't need anything else.

CONTROLLER SETUP:

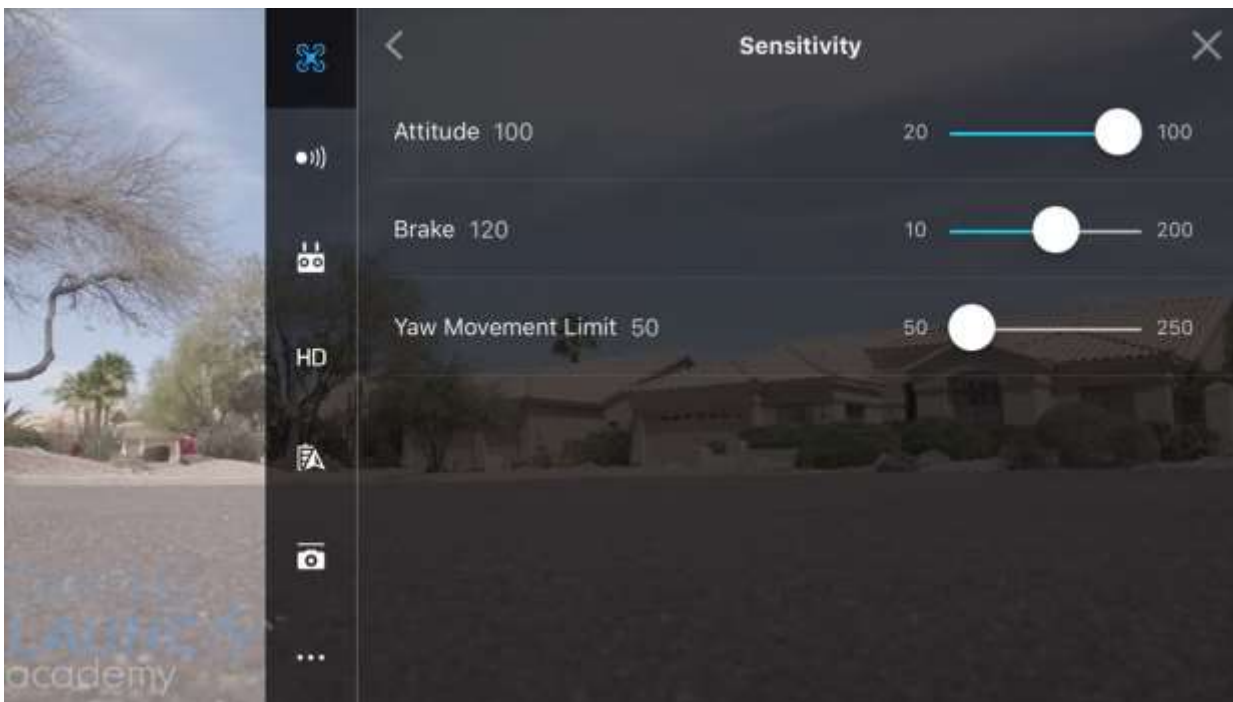
Here's my secret sauce. The default gimbal settings in all these Chinese drones are WAY to jerky, like a go-kart. We want the camera to flow and move smoothly. Start and stop softly. Any hard jerk makes your footage looking very human and it's distracting and doesn't meet broadcast standards.

Copy my settings.

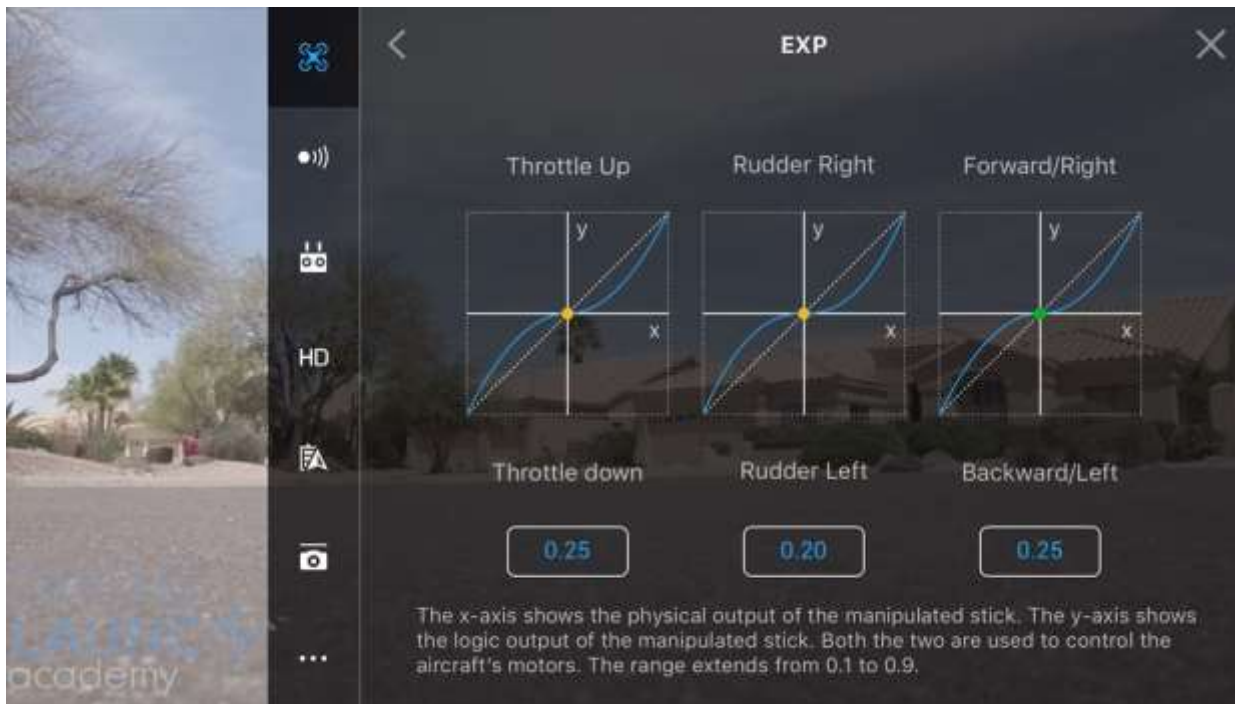
This is using the DJI app for any DJI drone.

It's in the **Settings (top right three little dots)** in the app while looking at your connected drone.

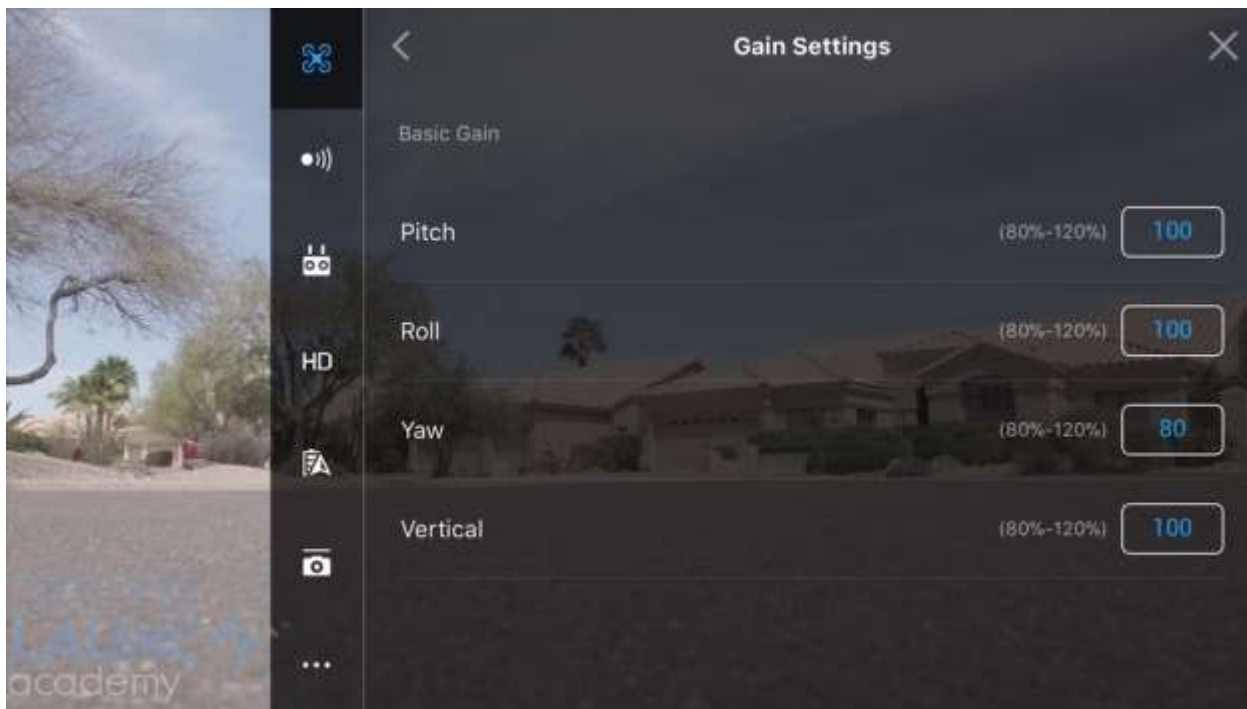
Click on the top Drone Icon > Advanced > Sensitivity. Copy these.



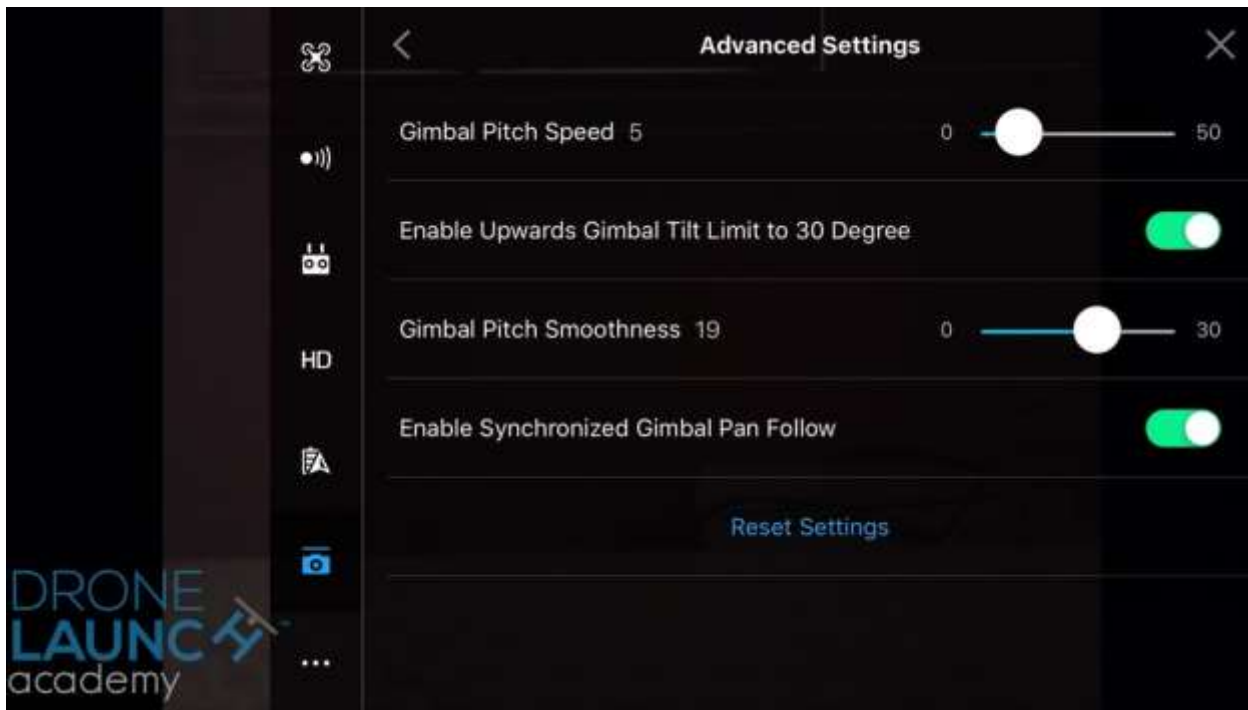
Then go back and choose EXP. Also copy these.



Now go back and choose GAIN SETTINGS



Finally, the gimbal. Choose the Camera Icon at the bottom left, and in its advanced settings, copy these. They may be too low for you, but I like it very slow.



It you're flying a Mavic 2 Pro FLY WITHOUT THE THUMB STICKS ATTACHED, it'll make it way simpler, the sticks are so annoying.

Now we move onto the actual image settings for the picture.

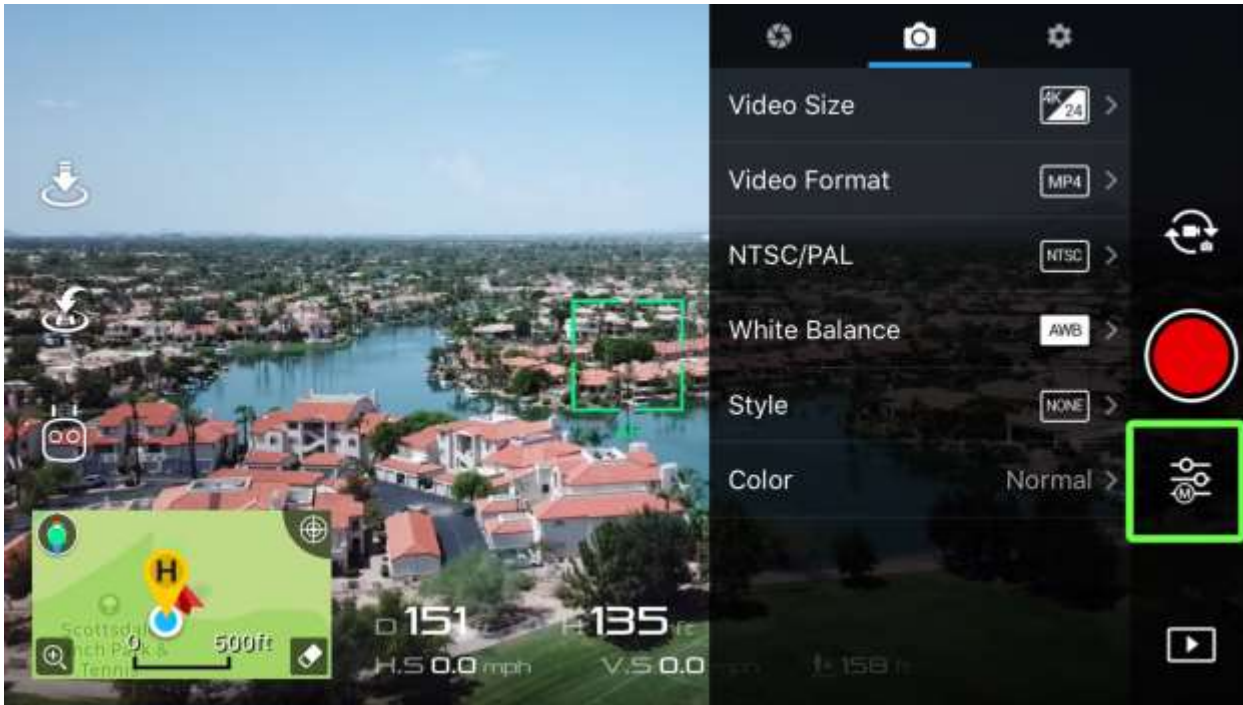


RECORDING SETTINGS:

Fly up into place, wherever you want to shoot.

NO auto-mode. ONLY shoot in manual.

Press the 3 bars to go into your picture setup.



VIDEO SIZE: 4K 24fps (movie frame rate) 3840x2160

VIDEO FORMAT: MOV

STYLE: (sharpness, contrast, saturation)

PHANTOMS -2, -3, -2

MAVICS: +1, -3, -2

Sharpness here is to fix some camera weirdness in both models manufactured settings.

Or too lazy to edit later? Want it straight from camera?

Phantoms -2, -1, 0

MAVICS: +1, -1, 0

(Mavic mini and Mavic Air 2 don't let you adjust this, honestly its okay they aren't those kinds of cameras)



COLOR:

Mavic Pro 1, Mavic 2 Zoom - D-Cinelike

Phantom 3 - Phantom 4 (not Pro) - D-Cinelike

Mavic Air - D-Cinelike, D-LOG if you are worried about shadows not being visible in broad daylight.

Phantom 4 Pro - D-Cinelike

Mavic 2 Pro – Normal... can't believe it, but yes normal.

Mavic Air 2 - No custom options here. It honestly is the first drone that looks okay by default, I would just shoot normal and skip the hassle. But D-Cinelike if you want to manually edit the color later.

*****In newer drones, Shooting MOV H.265 isn't playable on certain software that isn't super current and its advanced work in editing. If you are using software from this year you should be fine though.***

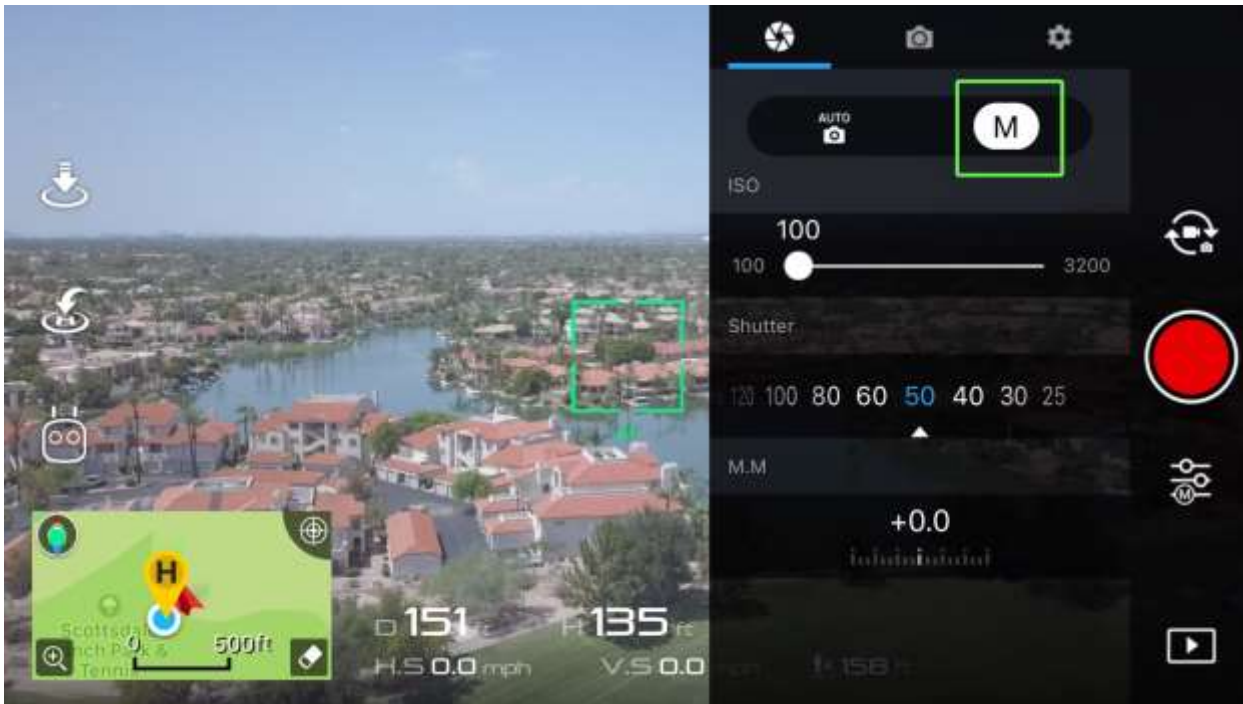
Otherwise H.264 is the backup plan.**

DJI drones have these Color Options. If you have a different brand, look for CINEMA or CINELIKE, or anything that looks really boring and colorless (but NOT BLACK AND WHITE)

SHOOT SETTINGS:

NO ISO. EVER. Leave it on 100, the lowest. (Phantom 4 Pro, maybe up to 800, but that's all.)

Remember we're in MANUAL mode, which is the big M.



Recommended – UHD 3840x2160 24fps, Shutter 50 (which means 1/50th, and you should aim to use double the frame rate (24fps) for your shutter speed)

If 4K 30fps - Shutter 60

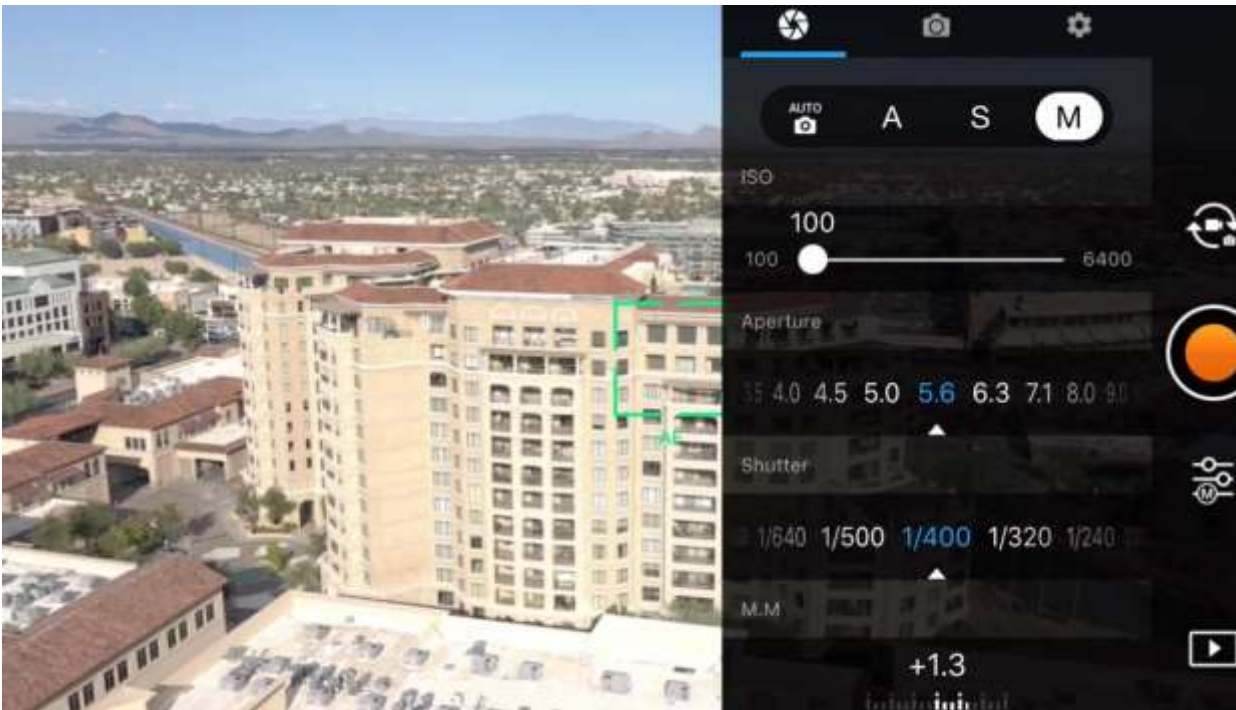
If 1080p at 120fps - Shutter 240 (Usually you shoot 120fps if you want slow motion later.)

This is where having an ND filter comes REALLY handy. Without one, you would have to shoot with a shutter of 1/1600 in broad daylight, which means the footage will be way too harsh and choppy looking with any movement, and not buttery smooth.

The bottom meter – the little M.M shows you the bright range the camera sees in front of it.

If it's at 0, it thinks things are level and lit correctly. If it were +1, you would be too high, and - .5 would be too dark. This is an auto reading and doesn't always know better, just use it to help guide you.

If you try to aim for the right shutter speed but the shot is looking overly bright - ALWAYS take the correctly-lit shot with higher shutter over having the correct 1/50th shutter but a very bright image. You won't really notice it that much if the shutter is like 1/100th or so, but it will help a TON if it means the shot isn't too bright now.



This shot is obviously overexposed. The building is what we are focused on and it fills most of the frame. The Exposure meter says 1.3 and the video is centered on the building , so we know that's where that reading is coming from. The REST of the image, like the mountains and trees, are evenly lit though. This would be a personal choice, but I would drop the shutter some to find a nice half -way point between the two.

APERTURE:

If you have the option (Phantom 4 Pro, Mavic 2 Pro) you can just lower the aperture to help you balance out the light, instead of changing the shutter. Anything between 2.8-8 is good, 11 or above might give you a teeny tiny blur, but is mostly not-noticeable.

Otherwise Aperture is too much to explain right now.

GENERAL TIPS:

I have a cinematography tips, posts, stuff everywhere you can learn about framing up a shot and what source material looks really good for filming.

This guide was just for setup. If you'd like to learn all those other things, here is my Webinar link:

<https://event.webinarjam.com/register/1/zn29lhx>

Or use **this** link to get **20% OFF the Cinematography and Editing course for drone video production**

This link also includes my Toolkit when you buy the course (Transitions, Titles, Graphic Effects, flares, etc.)

It's called **Aerial Video A to Z**. It will teach you everything you need to know:

- Coloring, and then coloring the *right* way
- Setup for Editing 4K footage at lightning speed, even on an old PC
- Tasteful Titles
- Instant fix to Shakey footage
- Fixing oddly blue tinted footage, also too bright/dark footage
- More cinematography lessons
- Equipment and Drone comparison/review
- So much more

With 320+ minutes of video, animated GIFs inside the text pages to make learning it all REALLY easy.



Hope this helped!



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